Universiteit Antwerpen

Non-viral CRISPR-engineering of CAR T cells: Where are we?

Faculteit Farmaceutische, Biomedische en Diergeneeskundige wetenschappen
Campus Drie Eiken

Simon Jacobs, Willems Hendrik
Flumens Donovan (co-promotor), Lion Eva (promotor)

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Jacobs Simon, Willems Hendrik

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Chimeric antigen receptor (CAR) T cells are a form of adoptive T-cell therapy widely investigated for the treatment of cancer. While viral transfection methods are the gold standard to generate CAR T cells, non-viral clustered regularly interspaced short palindromic repeat (CRISPR)-engineering using electroporation can create a more precise and homogenic form of gene-editing. This is accomplished by specifically cutting the DNA using a CRISPR-associated protein 9 (Cas9) and knocking-in a CAR transgene by stimulating with homology directed repair mechanism (HDR) with a HDR template. Moreover, non-viral CRISPR-engineering is faster, safer and cheaper than viral methods. Non-viral techniques for CAR T-cell engineering open a new and promising world in gene editing and immune therapy, with several early phase clinical trials in the pipeline.

Introduction:

Adoptive T-cell therapy, T cells and TCR

Over the past years immunotherapy has built up a track record in a broad range of diseases including various types of cancer, auto-immune diseases and infectious diseases (Sim et al., 2020, Tang et al., 2004, Krebs et al., 2013). One form of immunotherapy, that is widely investigated, is adoptive T-cell therapy. In those therapies the patient's own T cells will be isolated, ex vivo expanded and engineered to

improve the immune properties of the lymphocytes and redirect them to a specific disease-related antigen. This can be realised by genetically engineering T lymphocytes with transgenic antigen-specific receptors, such as T-cell receptors (TCRs) or chimeric antigen receptors (CARs). Subsequently, the engineered T cells will be administered back into the patient, making it a personalised therapy (Jafferji and Yang, 2019, Robbins et al., 2015). T-cell therapy is not limited to the utilisation of one specific type of T cell. State-of-the-art literature illustrates that, although they have a different mode of action, both $CD4^+$ T cells and $CD8^+$ T cells are used in T-cell therapy. $CD4^+$ T cells will act by activating and stimulating other components of the immune system. On the other hand $CD8^+$ T cells will be able to recognize and kill their target cell because of their cytotoxic properties (Yang et al., 2020). Naturally, T cells recognise a tumour-associated antigen (TAA) by their TCRs. The TCR is a heterodimer protein with an α and β chain. Both chains consist of a variable and a constant domain. Each variable domain is formed by recombination of the V, (D) and J segments, resulting in a hypervariable repertoire of TCRs. The specificity of the receptors is determined by the third complementarity-determining region (CDR3) loops on the variable domain. Each of the produced TCRs will be able to recognise a specific antigen, which must be presented by a major histocompatibility complex (MHC) molecule on the membrane of the target cells (Robins et al., 2009). This presentation of the antigen by MHC is being preceded by intracellular processing of the antigen by the antigen presenting cell (Blees et al., 2017). Even though this theory behind adoptive T-cell therapy with usage of naturally occurring TCRs is encouraging by itself, the therapy is associated with challenges. Cancer cells may be able to escape the immune system. One way they do this, is by minimising the amount of presented antigens. The cancer cells realise this by lowering the expression of MHC molecules on the plasma membrane (Fang et al., 2021).

Chimeric Antigen Receptors (CARs)

CAR T cells are T cells engineered to express a chimeric antigen receptor on their cell surface. CARs are artificial receptors consisting of an antigen recognition domain (ARD), a hinge domain (HD), transmembrane domain (TMD) and an intracellular signal transduction domain (STD). The ARD will be responsible for antigen recognition and will define the specificity of the CAR T cell. To construct the ARD, a single chain variable fragment of a monoclonal antibody directed against the antigen of interest is used. The HD and TMD have been shown to regulate the avidity and activity of CARs. Thus, it regulates the amount of signalling in T cells. To construct the TMD of CARs, CD8 or CD28 sequences are commonly used, because of their efficient expression on T cells. The STD's role is to start the intracellular signalling pathways after a conformation change of the STD because of the binding of the antigen to the ARD. The effectiveness of the signalling function depends on the composition of the STD. STDs always contain a CD3ζ part because it's needed to initiate the T cell activation (Fujiwara et al., 2020, Fujiwara et al., 2021, Hudecek et al., 2013). To extract the maximal potential out of the CAR signalling the need for co-stimulatory molecules is not to be neglected. This is why the current generation of CAR T cells possess costimulatory endodomains in their STD region, which will support the T cell activation. Some examples of these endodomains are following complexes: CD28, 4-1BB and OX40 (Savoldo et al., 2011). Unlike the TCR, the CAR is able to directly recognise extracellular antigens like CD19 or BCMA, without the need of intracellular processing and an MHC molecule to present the antigen (Roex et al., 2020, Fujiwara et al., 2020). CAR therapy shows promising results in haematological cancers but the results in solid tumours are up to now not so successful. The reason why CAR T cells are less effective in solid tumours may be the presence of an immunosuppressive tumour micro-environment helping the malignant cells by evading the immune cells. By combining CAR T therapy with other cancer treatments, such as combination

therapies with chemotherapeutics, researchers hope to improve the efficiency of the CAR therapy (Kuhn et al., 2019, Fujiwara et al., 2020, Srivastava et al., 2021, Chang et al., 2015).

CAR delivery systems

Transgene receptors can be introduced into an immune cell with viral or non-viral methods. Viral delivery systems are stable gene integration methods based on viruses to package and deliver transgenes and insert them into the hosts genome. Retroviruses (RNA), like lentiviruses and γ-retroviruses, use viral reverse transcriptase and integrase to (semi-)randomly insert sequences into a host cell (Milone and O'Doherty, 2018). However, because of this (semi-)random process there is a high chance of mutagenesis. Adeno-associated viruses (DNA) can counter this effect by inserting at a specific location, namely the AAVS1, on chromosome 19 (Wang et al., 2019). Over the past years various non-viral techniques were established to introduce transgene receptors in immune cells. Techniques that facilitate intake of proteins and nucleic acids, like lipid-based delivery (Li et al., 2021). Some studies use lipid-nanoparticles to deliver CAR mRNA in T cells, that results in transient expression of the CAR (Billingsley et al., 2020, Zhou et al., 2022, Parayath et al., 2020). Another study by Xu et al., 2019, uses cationic lipid nanoparticles to deliver pCas9 and sgRNA into cells (Xu et al., 2019). This means lipid-based delivery can be used to insert transgenes into the genome. The study of Parayath et al., 2020, could transfect the CAR into T cells using a transposon/transposase system. These systems (Piggybac/Sleeping beauty) use transposons containing the transgene between two terminal inverted repeats, that can stably integrate itself (semi-)randomly into the genome with the use of transiently expressed transposase (Kebriaei et al., 2017). This is great way to insert genes however, the non-random and precise process of CRISPR/Cas9 is more interesting for gene editing. But fundamentally, techniques that facilitate intake create transiently express genes. Like viral methods, they rely on particles that deliver the molecules. This subsequently raises the production cost, it is however simple to produce on a large scale (Priyadarshani, 2022). Alternatively,

physical methods like micro-injection or electroporation try to bypass the lipid membrane by going through it or increasing the permeability respectively. Electroporation is a wildly used system, because of its high efficiency and effectiveness compared to other techniques (Heller et al., 2008, Elmer, 2020). It uses electrodes to create short electrical pulses to increase the permeability of the cell membrane, so that nanopores appear and extracellular molecules can go through (**Figure 1**). In contrast to viral techniques larger molecules can be transfected (Elmer, 2020). However, electroporation has its disadvantages in the form of cytotoxicity. Higher voltages or longer pulses can make the permeability of the membrane irreversible or create a heath shock that can cause cell death (Tracy and Cadeddu, 2020, Kang et al., 2013). All the non-viral delivery techniques explained above provide transient expression of genes. But when combined with gene editing, cells that permanently express certain genes can be created. Recently, the field of electroporation and CRISPR-Cas9 merged to form a non-viral CRISPR-Cas9-based method to integrate transgenes in a specific site of the genome.

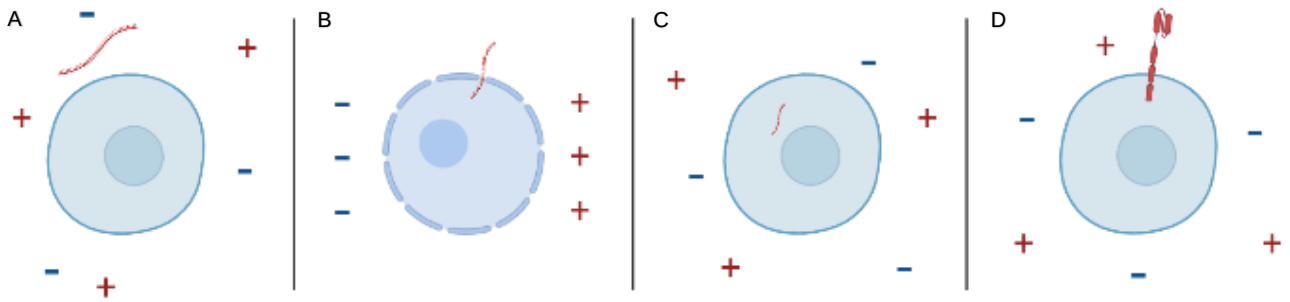


Figure 1: Simplified electroporation system: (A) Cells are placed between two opposite electrodes. (B) When these electrodes create an electrical field, the lipid bilayer will temporarily be permeable for nucleic acids/proteins by creating pores. (C) When the charge is turned off, the pores will close. (D) The inserted molecule can be used or expressed by the cell. (Created with BioRender.com)

CRISPR-Cas9-mediated gene editing

The clustered regularly interspaced short palindromic repeats (CRISPR)-associated protein 9 (Cas9) system has revolutionized medicine. Capable of precisely targeting and modifying genomic sequences,

this technology can be exploited for knocking out genes and introducing site-specific knock ins. Guided by a single guide RNA (sgRNA), Cas9 will specifically bind on a complementary target DNA sequence and create a double-stranded break (DSB) (**Figure 2**). Nuclease-induced DNA DSBs drive the activation of cellular DNA repair mechanisms. Gene knock-out is a result of non-homologous end joining (NHEJ) after a DNA DSB. This repair mechanism is prone to indel mutations and frameshift mutations resulting in the inactivation of the gene and thus knock-out of the gene. Gene insertions, on the other hand, can be mediated by exploiting the repair mechanism of homology-directed repair (HDR) after a DNA DSB (Ran et al., 2013). Compared to NHEJ, HDR requires the presence of a donor sequence homologous to the targeted chromosome site to guide repair of the DSB.

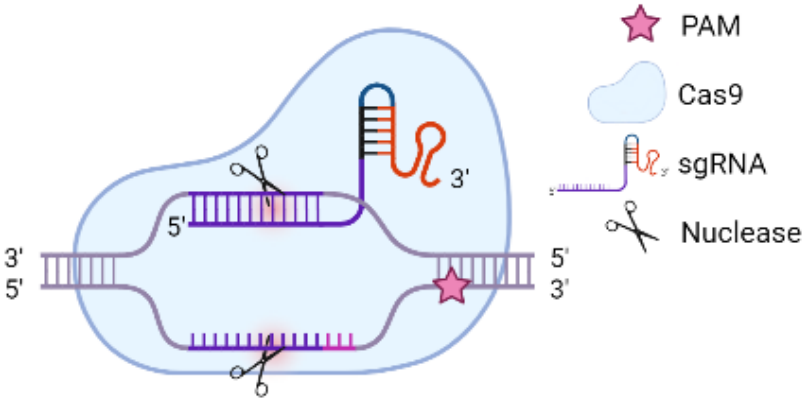


Figure 2: Cas9 protein binds on a target DNA sequence guided by a sgRNA, upstream of its binding signal (protospacer adjacent motif/PAM). Using nucleases, the complementary strand and the non-complementary strand will be cleaved. The cell will try and fix the DSB using NHEJ, which is prone to indel mutations and a consequent knock-out. Adding a transgene with the right homology arms, complementary to the breakpoints, HDR can be performed. This will lead to a knock-in of the transgene and knock-out of the cleaved gene. (Created with BioRender.com)

It is the mechanism of HDR that forms the basis of non-viral CRISPR/Cas9-mediated engineering of CAR T cells. By designing a specific HDR template for a gene of interest, DNA polymerases have a template to restore the DSB and replicate the gene of interest into a defined locus of the genome. This template has homology arms on both ends complementary to the breakpoints, so that it can function as a homolog in

HDR. HDR templates can be double-stranded DNA (dsDNA) or single-stranded DNA (ssDNA). HDR templates consisting of dsDNA can contain longer sequences as opposed to ssDNA templates. However, the use of high concentrations of dsDNA is toxic for the cell. dsDNA HDR templates are also characterized by a lower HDR efficiency compared to their ssDNA HDR template analogue (Miura et al., 2015, Shy et al., 2022). The higher HDR efficiency of ssDNA HDR templates is related to the synthesis-dependant single-strand annealing. It is however more difficult to create stable long sequences of ssDNA HDR templates (Miura et al., 2015, Shy et al., 2022). Generally, an HDR template is constructed by different elements. First of all, there is a left and right homology arm homologous to the target site of insertion. Next to the left homology arm -downstream from the transgene- a porcine teschovirus-1 2A self-cleaving peptide (P2A) sequence can be found. This sequence codes for a self-cleaving 2A protein, which cleaves itself, the left homology arm and the residue of the old gene from the transgene during translation. The last elements are the transgene followed by a stop codon (**Figure 3**). The transgene will be kept under the control of the endogenous promotor. Selection of the target locus in the genome depends on what gene will be knocked out after insertion. This means that the knock-out cannot be detrimental for the T-cell and its function. Interesting target genes described in literature are the T-cell receptor constant α-chain and β-chain, TRAC and TRBC respectively (Eyquem et al., 2017), programmed cell death genes (PD-1) (Su et al., 2016) or adeno-associated virus integration site 1 (AAVS1) (Chen et al., 2020).



Figure 3: HDR template consisting of the two homology arms, the cleavable p2A sequence, the transgene and a stopcodon. The transgene will be kept under the control of the endogenous promotor. (Created with BioRender.com)

Non-viral CRISPR-engineered CAR T cells in clinical trials

Since non-viral CRISPR-engineering is an emerging method of creating CAR T cells, only a few early-stage clinical trials were available. Some active trials were found that use a non-viral delivery strategy for the introduction of the Cas9-complex and a HDR template. An active (not recruiting) trial (**NCT04213469**) uses PD1-CD1-CAR T cells in patients with relapse/refractory B-cell Non-Hodgekin Lymphoma (r/r B-NHL) sponsored by Bioray laboratories. In this study, T cells are engineered with a CD19-CAR dsDNA HDR inserted in the PDCD1 locus, and thus express the T cells a CD19-specific CAR while simultaneously knocking-out PD-1 (Zhang et al., 2022). Another recruiting trial (**NCT04637763**) is using CB-010 in patients with r/r B-NHL, a product developed by Caribou Biosciences, Inc.. CB-010 is an anti-CD19 CAR T cell therapy where a ssDNA CAR HDR template is inserted into the TRAC locus, consequently knocking out the native TCR. A spCas9 with hybrid RNA-DNA (chRDNA) guides is used (Donohoue et al., 2021). Caribou Biosciences uses CB-010 in another trial (**NCT05722418**), were it targets B cell maturation antigen (BLMA) in r/r multiple myeloma (r/r MM).

Table I: Clinical Trials

	Disease	**CAR target antigen**	**Homology Directed Repair**			**Electroporation**		**Cas9 format**	**Clinical Trial Identifier and References**
			Target gene	**ssDNA/dsDNA template**	**Enhancing molecules**	**Electro-porator**	**Settings**		
1.	r/r B-NHL	CD19	PD-1	dsDNA	/	Lonza 4D electro-poration system	Programme EO115	Recombinant spCas9	NCT04213469
2.	r/r B-NHL	CD19	TRAC	ssDNA	/	Lonza 4D 96-well format Nucleofector	Pulse code: CA137	spCas9	NCT04637763
3.	r/r MM	BLMA	TRAC	ssDNA	/	Lonza 4D 96-well format Nucleofector	Pulse code: CA137	spCas9	NCT05722418

Conclusion

Adoptive CAR T-cell therapy turned out to be successful in combatting haematological cancers. However, it is still awaiting positive outcomes in trials with solid malignancies (Kuhn et al., 2019). While most CAR T-cell therapies rely on viral delivery methods of the CAR, non-viral techniques are increasing in popularity. Nevertheless, non-viral CAR T-cell therapies in clinical trials are recently being used (**NCT04213469**, **NCT04637763**, **NCT05722418**). Although these methods lose the viral insertion and delivery, non-viral state-of-the art delivery systems can transfer nucleic acids and proteins into the cell with high efficiency and safety (Heller et al., 2008, Elmer, 2020). Electroporation has one of the highest efficiency and effectiveness, because of its ability to transfer large molecules without the use of delivery-particles (Elmer, 2020). The CRISPR/Cas9 system can replace the random viral transduction of transgenes with remarkable precision. Because of this, CRISPR/Cas9 is a safer insertion method compared to viral vectors that insert sequences randomly, which can lead to insertional mutagenesis (Milone and O'Doherty, 2018, Ran et al., 2013). Delivering the CRISPR/Cas9 components using electroporation greatly advanced non-viral engineering in gene editing. In conclusion, non-viral CRISPR-engineering of CAR T cells is a young but promising new adoptive T-cell therapy.

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Internship assay

2D3 cell line growth and stimulation with ionomycin and phorbol 12-myristate 13-acetate

Objectives

In this experiment the expansion and the non-specific activation potential of TCRαβ-deficient, CD8αβ and NFAT-EGFP stably-transfected T cell acute leukemia 2D3 cell line activation was evaluated. Therefore, 2D3 cells were cultured for 3 weeks with viability assessment every three to four days. In addition, 2D3 cells were activated in a non-specific manner using PMA and ionomycin. Resulting EGFP expression was followed up over time.

Materials and methods

2D3 cell line

The 2D3 cell line is a TCRαβ-deficient, CD8αβ and NFAT-EGFP stably-transfected T cell originating from an acute leukemia cell line. Characterized by the lack of a natural TCR, and a nuclear factor of activated T cells (NFAT) promoted enhanced green fluorescent protein (EGFP) reporter gene, makes the 2D3 cell line an interesting model for T cell activation. Upon activation of the cell line, EGFP will be expressed and can be measured via flow cytometry. 2D3 cells can be used to evaluate the T-cell activation following siRNA-silencing (van Ens et al., 2020), tumor-specific T-cell receptor (TCR) expression (Campillo-Davo et al., 2020) and so on.

Cell counting and viability test

2D3 cell concentration was determined using an ABX Micros EZ (HORIBA) automatic cell counter. The viability of the cell line was assessed with a propidium iodide (PI; Invitrogen) staining and measured on a

CytoFLEX (Berkman coulter) in the PerCP-Cy5.5 channel (blue laser: 488nm). Propidium iodide binds to the DNA of dead cells.

Thawing 2D3 cells

Cryopreserved 2D3 cells ware rapidly thawed by briefly (20s) heating up the cryotube in a prewarmed (37°C) warm water bath. Subsequently, the cells are transferred to preheated recovery medium (Iscove Modified Dulbecco Media (IMDM) + 10%Fetal Bovine Serum (FBS)) of 37°C to dilute dimethyl sulfoxide (DMSO). DMSO is a toxic cryoprotectant which will prevent the formation of ice crystals when cryopreserving immune cells. After thawing, 2D3 cells rested for 15 minutes at 37°C and 5% CO_2 in an incubator. Next, cells were washed and resuspended in 10 mL of culture medium (IMDM + 10% FBS) and transferred to a 25 mL culture flask. The next day, cell concentration and viability were determined and cells were set on a concentration of $0.3*10^6$cells/mL to start the cell culture.

Cell line culture

IMDM medium contains phenol red, a red indicator that turns yellow when the solution is acidic. The yellow is an indication that the medium must be refreshed, and the cells should be reconcentrated to a lower concentration. Each passage, 2D3 cell concentration and viability was determined. Subsequently, cells were diluted with culture medium (IMDM + 10%FBS) in an end volume of 10 mL and a concentration of 0.3×10^6 cells/mL. Excess 2D3 cells were discarded. Passaging the cells was repeated every three to four days.

Stimulation of 2D3 cells

2D3 cells were stimulated for 24 hours using PMA and ionomycin. To stimulate the 2D3 cells, cells were washed and resuspended in activating medium consisting of IMDM + 10%FBS + PMA [2µg/mL] + ION

[1µg/mL]), considering that there is a 2:1 ratio between PMA and ION. An unstimulated control was included, which was cultured in IMDM + 10%FBS.

EGFP expression test

EGFP expression was measured using the Cytoflex (Berkman coulter) on the fluorescein isothiocyanate (FITC) channel (blue laser: 488nm). During the first 4 hours a measurement was performed for every stimulated sample alongside the negative control every hour. The next day the last timepoint was measured at 24hours after stimulation.

Data analysis

FlowJo™ Software was used to analyze flow cytometric data GraphPad PRISM was then used to graphically visualize viability, normalized fold increase and EGFP expression.

The fold increase (FI) is calculated and used to compare the evolution of cell growth during cell culture. It is the ratio between the concentration at the end of the expansion and the starting concentration ($0.3*10^6$cells/mL). By normalizing the fold increase (FI°) per day (n), taking the estimated 24h doubling time of Jurkat T cells into account as an exponential variable, a relevant formula can be constructed: $FI° = \frac{FI}{2^n}$ (Schoene and Kamara, 1999). This means that generally, not considering the cell-environment interactions, the FI° should be equal to 100%. Although this can be used as a guide to compare cell culture, there are other factors (medium, pipet precession, interaction, type of cell etc.) that influence growth. To assess the upregulation of EGFP by activated 2D3, statistical analysis was performed by comparing the unstimulated cells and the stimulated cells after 24 hours. This was accomplished by using an unpaired T-test. P values <0.05 were deemed significant.

Results

Stable Cell growth and viability

During the 2D3 cell line culture, viability remained stable over time (**Figure 1**). It started off with a viability of 86% and was followed by a minor decrease on the seventh day (84%), but quickly stabilised. The highest viability that occurred was 89% on day 15. Overall, it stays consistent around the mean of 86% with a standard deviation of zero. When looking at the concentration after every passage, it varied remarkably between a minimum of $0.8*10^6$cells/mL and a maximum of $2.0*10^6$cells/mL. This cell line should almost double in amount every other day. However, this is not the case, even after not considering the cell-environment interactions that slow proliferation. The FI° visualises a clear evolution of cell growth because it is normalized for each day (**Figure 2**). The first cultivation has a 50% increase, but the second cultivation only shows an 18% increase. The last cultivations seem to stabilize in growth due to the consistent increase of about 40%.

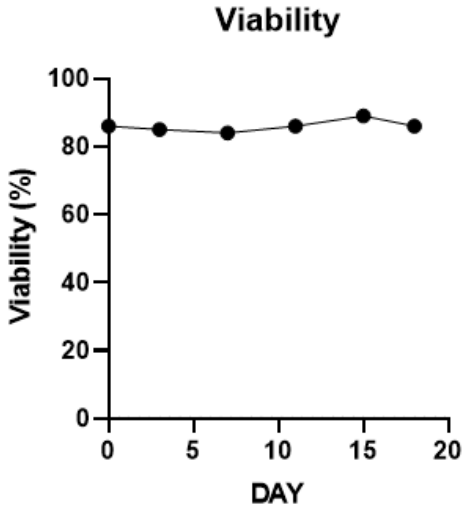


Figure 1: The viability of the 2D3 cell line during culture. Viability of the 2D3 cells was measured each three to four days for 15 days using PI. It started off with a viability of 86%. There were some minor decreases, like on day 7 (84%), and minor increases, like on day 15 (89%). In the end the viability was stable, with an average of 86% and had a standard deviation of zero.

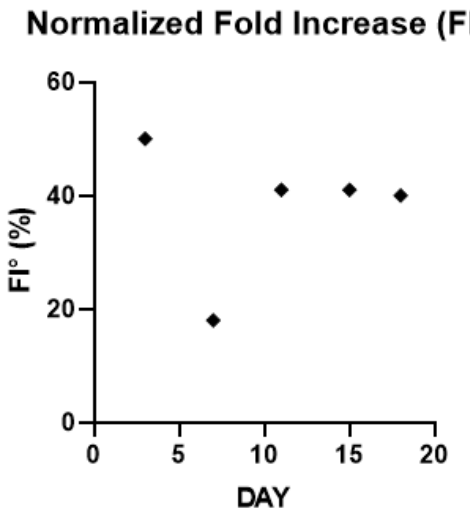


Figure 2: The fold increase between two consecutive passages normalized for each day. The normalized fold increase is the ratio between the end concentration and starting concentration of each passage for each day, considering the exponential growth of 2D3 cells. The first cultivation has a high FI° but is immediately followed by a steep decrease. Growth of the final cultivations on day 11, 15 and 18 indicate a stabilisation over time.

Rising EGFP expression overtime in Ionomycin and PMA stimulated 2D3 cell line

After the first gating step of the flowcytometry data an estimation of the viability is obtained (**Figure 3**). The selected population is seen to be decreasing in the stimulated samples. At the 24h timepoint only a small portion of the initially gated population remains, while all the other events are dead cells or debris. Looking at the graph in **figure 4** visual conformation can be done to report a difference between the evolutions of expression profiles of the stimulated and unstimulated cells. An increase in expression between successive timepoints can be noticed by looking at the graph of the stimulated samples. According to the statistical analysis there is a significant difference between the EGFP expressions in the unstimulated and stimulated cells after 24 hours. The difference in EGFP is also visible on the flowcytometry plots in **figure 3b** and **3c**. While the expressed EGFP rises over time, the amount of EGFP expressing cells remains at a stable baseline, which never crosses 0.78% stimulated cells in any single measurement. In addition, the single measurements variate more within one timepoint of the stimulated cells compared to the non-stimulated.

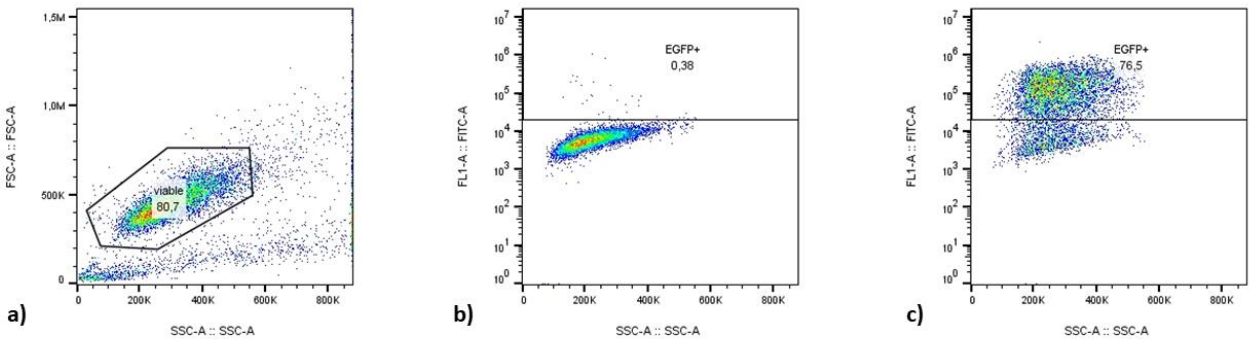


Figure 3: Representative illustration of the gating strategy used to identify EGFP positive cells. To illustrate viability, forward scatter (FSC-A) is plotted against side scatter (SSC-A). To select the viable cells, gate around the population consisting of the larger (high FSC) and a rather low internal complexity and smooth cell membrane (low SSC). To gate for the EGFP+ cells establish a gate above the population present in the negative control when FITC-A is plotted against SSC-A. Once the expression of EGFP initiates the cells will show a higher FITC signal, which will integrate them in the EGFP+ gate. **3a** represents all the events observed by the flow cytometer by plotting FSC-A against SSC-A. The performed gate will separate the 2D3 cells from the dead cells and debris by estimation. **3b** is an example of a non-stimulated sample comparing FITC-A with SSC-A. this negative control was used to set the threshold between cells expressing EGFP and cells who do not express EGFP. **3c** represents a stimulated sample after 4 hours using the same parameters as 3b.

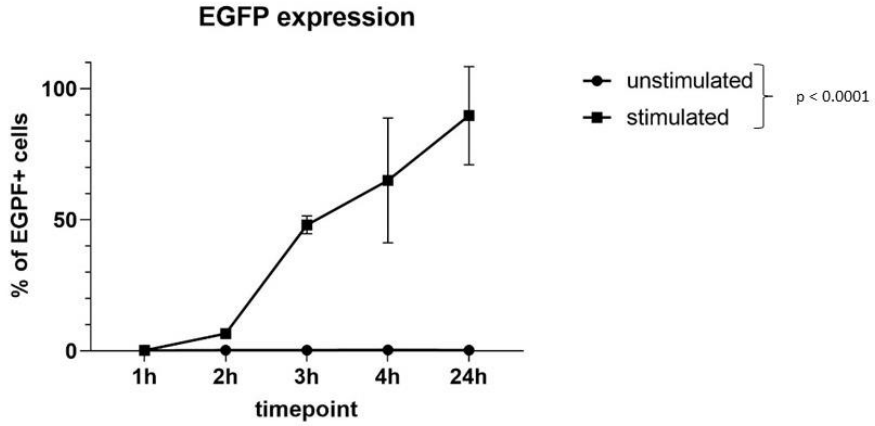


Figure 4: EGFP expression (%) for every measured timepoint, comparing stimulated with unstimulated samples. The percentage of EGFP positive cells is defined as the population of cells. Every timepoint on the graph represents the means of these expression percentages for the separate populations.This percentage seems to rise over time in the stimulated population, while it remains stable on a low level in the unstimulated cells. On the 4h and 24h timepoints the standard deviations in the stimulated populations were respectively 23.750 and 18.701 while for non-stimulated populations these timepoints had SDs of 0.213 and 0.189. The 2 large error flags can be explained according to these high SDs because of the occurrence of a radical outlier in both the 4h and 24h timepoints. The p value obtained after performing an unpaired T-test comparing the means of the 24h timepoints of unstimulated and stimulated populations was smaller than 0.0001. This indicated a statistically significant difference in the means of the unstimulated and stimulated cells. While the mean of every timepoint of the unstimulated cells remains beneath 0.4% the means of the stimulated cells increase from 0.200% at 1h after stimulation to 6.648% at 2h, 48.075% after 3h, 65.044% after 4h and 89.750% at the last timepoint after 24h.

Conclusion

The 2D3 cell line proves itself to be a notable experimental cell line, because of its fast growth rate and ability to show activation. The results of the viability and normalized fold increase indicate that the 2D3 cell line is an optimal model for usage in experiments because of their consistently high viability and relatively rapid growth. Between passages of the cell line the viability shows to remain stable. The start of the culture had a fast proliferation and the growth of the final flasks stabilized over time. Because Jurkat T cells have

a doubling time of 20.7 ±2.2 h, which can be extrapolated to a doubling time of one day (Schoene and Kamara, 1999), the daily growth could be expressed as a normalized fold increase (FI°), where 100% FI° corresponds to a doubling time of 24h. However, other variables (medium, pipet precession, interaction, type of cell etc.) were not included in this calculation, this means that a FI° between 40% and 50% could be standard for this cell line. On day seven, there was a drop in growth. This is in all probability due to splitting of the cell line in multiple flasks for experimental purposes. This sequence of actions may have put the cells under more stress than usual and may have slowed down the proliferation. The continuous cell line displays rapid proliferation with minimal loss of cells. This makes it a cost-effective and suitable cell line for various experiments regarding innovating T-cell research. The 2D3 cell line contains NFAT-EGFP stably-transfected cells. This means that an insertion of the EGFP gene and viral promotor of NFAT was performed. Activation of the cell and consequently NFAT will start expression of measurable EGFP. However, 2D3 cells are TCRαβ-deficient and can only be activated synthetically. PMA and ION (PMA/ION) were used for non-specific activation. ION is an ionophore that will increase calcium influx. Calcium, together with calmodulin will activate phosphatase calcineurin that in turn will activate NFAT and consequently EGFP expression. PMA has a synergic effect by stimulating protein kinase C, that will stimulate activation pathways (Li and Costa, 2022, Chatila et al., 1989). PMA/ION need time to activate T cells and in doing so, some cells undergo activation induced cell death. 2D3 cell activation was determined by measuring the EGFP expression levels over time. After 24h stimulation, a maximal amount of EGFP expressing cells (89.750%) was reached. In comparison to the stimulated cells, the unstimulated cells did not show EGFP expression (only 0.281% after 24h) as a negative control. The usage of PMA/ION and its activating effect indicate that it can be used as a useful positive control. Overall, when rapidly screening TCR or CAR transgenes, the 2D3 cell line is versatile and effective T cell model (Campillo-Davo et al., 2020).

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